Iván Rodrigo Pianto Antón

Full stack software engineer · AI systems architect · Tech lead

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PROFILE

I build software for real operations: multi-tenant ERPs, e-commerce platforms run by AI agents, civil-engineering tools and hydrological risk models. I combine a civil-engineering background with full stack architecture, and I work with discipline: specs before code, tests first, adversarial review and honest metrics.

Key stack: TypeScript, Python, Next.js 16, NestJS, FastAPI, PostgreSQL

Key figures: 13,223 tests in the release gate of a multi-tenant ERP; 20,491 ads analyzed with AI for USD 25.53; 147k stamps in a catalog rebuilt by scraping; 25,290 river reaches in a national flood-hazard run.

EXPERIENCE

Tech lead and systems & AI architect — Peruvian software and AI company

2024 – present

Architecture and development of a multi-tenant traceability and foreign-trade ERP in production, automated with AI agents.

Lead full stack developer — Latin American e-commerce startup

2024 – 2025

Frontend and backend architecture of an omnichannel CRM and ERP with WhatsApp messaging and payment gateways.

Open-source author and applied engineering — Independent

2025 – present

Tools for AI agents, civil-engineering software, and traffic and hydrology models.

SELECTED PROJECTS

Multi-tenant ERP and traceability for agro-export

Peruvian seafood exporter, through a software company · 2024 – present

- Full foreign-trade area: proformas, containers, automatic packing proposals, a Kanban board and a historical load of 240 proformas (~USD 49.9M).
- Release gate with 13,223 tests, Prometheus and Grafana observability, and infrastructure cut to ~USD 83/month.

E-commerce platform run by AI agents

Latin American e-commerce startup · 2024 – present

- v1: an 11-agent AI pipeline (DAG) with 2 human checkpoints that validates a product and produces avatar, competitors, branding, copy, creatives and a landing page in ~30–45 min.
- v2: rebuilt as a multi-tenant SaaS in 20+ chained changes: identity, credit-based billing, financial engine, landing pages, publishing, orders and shipments.

Niche radar: AI ad-market intelligence

Module of the e-commerce platform · 2026

- Parallel collector with 20–30 workers: 923–1,717 ads per minute versus 68 sequentially.
- 20,491 ads analyzed for USD 25.53 and 62,581 phrases clustered for USD 0.40.

Construction company platform: procurement, estimating and tenders

Construction contractor in northern Peru · 2026

- Internal platform with suppliers validated against a registry of 18.5M tax IDs, requests for quotation, price comparison and 13 corporate templates.
- Unit-price and quantity-takeoff engine with a hierarchical predictor: 13.8% median error on blind sets, over a corpus of 888 public works.

Road intelligence platform for Peru

Own project · 2026

- National traffic model (143k km network, 253 zones, ODME against 73 toll stations): typical travel-time error from 55.6% to 13.0%.
- Urban corridors for 12 cities calibrated to a 6.6% out-of-sample MAPE by day type.

Flood and debris-flow early warning

Own project, piloted in Piura · 2026

- Ensemble rainfall → LSTM discharge → Gumbel thresholds at 6 bridges: NSE 0.793 versus 0.597 for the baseline, and a Brier skill score of 0.60.
- Honest validation: the classic 20 mm rule had no skill (POD 0.03–0.16); the new debris-flow model reached TSS 0.81 on a locked 2025–26 set.

Live meeting copilot with on-device speech recognition

Own product · 2026

- Streaming ASR on CPU with a 0.18–0.21 real-time factor, validated on a real 2 h 23 min meeting.
- Found and fixed a use-after-free in the voice detector that corrupted 5.5% of timestamps.

Mobile cockpit for AI coding agents (open source)

Own open-source project · 2025 – present

- QR pairing with end-to-end encryption (Noise IKpsk2) and a containerized relay with self-hosted push.
- 2,600+ tests across packages (server 889, mobile 1,530, protocol 231).

SKILLS

Frontend and mobile: TypeScript, React 19, Next.js 16, Tailwind CSS 4, Expo, React Native, Vite, Alpine.js, MapLibre

Backend and APIs: Node.js, NestJS, Fastify, Express, Python, FastAPI, WebSockets, MCP, BullMQ, Google Apps Script

Data: PostgreSQL, SQLite, Cloudflare D1, Redis, Drizzle, SQLAlchemy, DuckDB, pandas, Excel · Power Query · VBA, Streamlit

AI and machine learning: Multi-agent orchestration, LLM document extraction, Speech to text (Whisper, ONNX), Embeddings and vector search, PyTorch, scikit-learn, Model evaluation

Cloud and DevOps: Docker, nginx, GitHub Actions, Cloudflare, Vercel, Google Cloud, systemd, Prometheus, Grafana, Linux

Quality: TDD, Vitest, Jest, pytest, Playwright, Adversarial review

Engineering and domain: Cost estimating (unit prices, takeoffs), Public procurement, Works control (earned value), Traffic modelling, Hydrology and early warning, Foreign trade, E-invoicing, Financial analysis, Game design

EDUCATION

- Civil Engineering and Information Technology — university education (Piura, Peru)

LANGUAGES

- Spanish — native
- English — professional technical